Indeterminism and Randomness in Quantum and Classical Physics

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The Principle of Sufficient Reason (PSR)

Leibniz's formulation (*Monadology* §32, 1714)

"*And that of sufficient reason, in virtue of which we hold that no fact can be real or existent, no statement true, unless there be a sufficient reason why it is so and not otherwise, although these reasons usually cannot be known by us.*"

— G. W. Leibniz, *Monadologie*, §32 (written in French; publ. posthumously)

How Leibniz meant it

- **Metaphysical, not formal:** an axiom about *reality* and its intelligibility — never stated in a calculus or derived from logic.
- **"Reason" = ground/explanation**, wider than *cause*: it answers *why so and not otherwise*.
- **Epistemically modest:** the sufficient reason often involves an infinite analysis — fully surveyable only by God.

Suppose the answer is "not real"— or inaccessible by the available means

1. **A Viennese passer-by:** "Where in Vienna is St. Paul's Cathedral?" The named cathedral has no Viennese referent; the famous one is in London. Vienna's central cathedral is St. Stephen's.
2. **A car:** "How much milk is in your refrigerator?" An ordinary car has no relevant sensor or data channel.
3. **A large language model:** Ask for a fact that is neither represented in its learned evidence nor supplied in the context or through tools.
4. **A quantum system:** Prepare normalized $|\psi\rangle$ and ask the yes/no question $P_\phi = |\phi\rangle\langle\phi|$, with $0 < |\langle\phi|\psi\rangle|^2 < 1$. The prepared state is not an eigenstate of this query.

What happens when an answer is enforced?

1. **The passer-by** may repair the false premise, ask for clarification, substitute St. Stephen's, or simply guess.
2. **The car** can return "unavailable", request access to a refrigerator sensor, or emit a value unsupported by its data.
3. **The LLM** may halucinate a plausible but ungrounded answer. Alignment and tools can encourage abstention or evidence-seeking, but cannot guarantee either.
4. **The quantum measurement** records 1 with $p = |\langle\phi|\psi\rangle|^2$ and 0 with $1 - p$. In the complementary qubit case, $p = 1/2$.

The contrast

The first three cases concern false premises, missing access, or unsupported inference. The fourth is a positive Born-rule prediction: an admissible outcome must occur, although the prepared state fixes only its probability.

Stace: realism exceeds possible experience

The claim he targets

Realism says that some entities sometimes exist without being experienced by any finite mind. Stace does *not* prove their nonexistence. His conclusion is epistemic: their unexperienced existence is neither known nor made probable.

Three routes fail

Perception: experiencing the unexperienced is contradictory.

Induction: no unexperienced existence has ever been observed.

Deduction: an object's absence during an observational gap is not logically inconsistent.

Two replies fail

Egocentric predicament: ignorance does not prove nonexistence, but it provides no reason for existence either.

Causal continuity: it presupposes that unobserved processes and laws persist — the very point to be proved.

Stace's conclusion

Persistence through "interperceptual intervals" is a simplifying mental construction or fiction, not an established observer-independent fact.

DOI sources: Stace 1934.

Quantum "anti-realism": localized indefiniteness

Theorem 2: a finite witness for every incompatible query

Let $n \geq 3$ and prepare a pure state $|\psi\rangle$, so the eigenstate assumption fixes $v(P_\psi) = 1$. For every unit $|\phi\rangle$ with $0 < |\langle\psi|\phi\rangle| < 1$, one can effectively construct a finite set O_ϕ of rank-one projectors containing P_ψ and P_ϕ for which no admissible noncontextual value assignment can make P_ϕ definite.

Angle from the prepared ray

Let $\theta = \arccos|\langle\psi|\phi\rangle|$, with $0 \leq \theta \leq \pi/2$.

- $\theta = 0$: same ray, $P_\phi = P_\psi$; definite value 1.
- $0 < \theta < \pi/2$: neither aligned nor orthogonal; P_ϕ is value indefinite in the theorem's sense.
- $\theta = \pi/2$: orthogonal ray; definite value 0 under admissibility.

What "anti-realism" means here

Almost every rank-one proposition lacks a pre-existing *noncontextual* 0/1 value relative to the prepared state. The result assumes the eigenstate condition and admissibility; each target P_ϕ has its own finite witness O_ϕ. It does not rule out contextual hidden-variable accounts or every form of metaphysical realism.

DOI sources: Abbott–Calude–Svozil 2015, Theorem 2.

Quantum indeterminacy in Zeilinger's account

Finite information

Zeilinger's foundational principle says that an *elementary* system represents the truth value of one proposition — one bit of information.

A complementary query

For a yes/no proposition not fixed by that bit, quantum mechanics supplies Born probabilities rather than a pre-existing eigenvalue. Zeilinger interprets the individual outcome as irreducibly random: the "message of the quantum".

PSR, Gaps and continuous creation

Forget about The Principle of Sufficient Reason—its false! Philipp Frank argued that in such cases mechanical laws may leave *gaps* (for *miracles*): an observed initial state need not determine a unique observable future. By analogy, putatively uncaused event actualization can be called *"creatio continua"*. In theology the term usually means ongoing creation or sustenance, not a physical randomness mechanism.

DOI sources: Zeilinger 1999; Zeilinger 2005; Frank 1998; Svozil 2016.

Quantum examples proposed for *creatio continua*

Candidate events

- **Spontaneous emission:** the registered decay time and emitted mode are probabilistic, although the closed atom–field state can evolve unitarily.
- **Stimulated emission:** an external field drives the transition; it can be coherent and unitary, so it is not by itself a collapse.
- **Query mismatch:** measuring P_ϕ on a non-eigenstate $|\psi\rangle$ produces a Born-distributed result.

Measurement dynamics — violation of unitary confinement

- **von Neumann Process 1 (violation of unitary state evolution):** projection/collapse — a nonunitary state update.
- **Process 2 (unitary confinement):** unitary, reversible evolution preserving inner products and distances.
- **Partial trace:** a reduced subsystem description, not a separate objective collapse.

Ways to cope?

- Everett retains universal Process 2 dynamics. Thus "creatio continua" belongs to collapse-based readings, not to Everett's fundamental dynamics.
- Infinite succession of Wigner's Friends, modelled by infinite tensor products, yields unitary non-equivalence.
- Lemmas by Lévy's and Johnson–Lindenstrauss on FAPP-orthogonality.

Kreisel and Specker: limits of computation

Kreisel's criterion

A theory is *mechanistic* if every observable sequence or real number it defines is recursive in the data that determine it. This computability criterion cuts across deterministic and probabilistic theories.

A recursive curve with no computable maximizer

There is a recursively continuous $f : [0, 1] \to \mathbb{R}$, with a recursive modulus of continuity, whose maximum is attained at no recursive x. The maximum value is the computable number $1/36$; its maximizing arguments are noncomputable.

A Specker sequence: computable terms, noncomputable limit

There is a computable, monotonically increasing, bounded sequence (q_n) of rational numbers whose limit $\alpha = \lim_n q_n$ is not a computable real. Consequently, the convergence has no computable rate: no computable $N(m)$ can guarantee $|q_n - \alpha| < 2^{-m}$ for every $n \geq N(m)$. Every finite term can be generated effectively; the asymptotic value cannot.

DOI sources: Kreisel 1974; Specker 1949; Specker 1959 (1990 reprint).

Classical indeterminism and randomness

Continuum and chaos

- $\mathbb{R}$ is uncountable.
- Lebesgue-almost every real is Martin-Löf random; choice alone supplies neither sampling nor randomness.
- Chaos can amplify small uncertainty; injectivity is unnecessary.

Beyond ordinary computation

- The diagonal Ackermann function is total and computable, but not primitive recursive.
- Chaitin's Ω_U is a machine-dependent, uncomputable, Martin-Löf-random *real* — not an evolution function.
- The busy-beaver function $\Sigma(n)$ is total and uncomputable; it eventually exceeds every total computable function.
- A partial f is undefined on some inputs: that is failure or nontermination, not stochastic choice.

DOI sources: Martin-Löf 1966; Ackermann 1928; Chaitin 2007; Radó 1962.

Vacuous "laws" and correlations through Ramsey's theorem

Ramsey's theorem

Fix positive integers k, r, c. For all sufficiently large N, every c-colouring of the k-element subsets of an N-element set contains an r-element subset whose k-element subsets all have the same colour.

Some regularity is unavoidable

Encode a finitely classified relation among k observations as a colour. In a sufficiently large data set, one can then select a homogeneous substructure: within that selected set, the same relation holds throughout. Thus an apparently striking correlation may be guaranteed to occur *somewhere*, even when no dynamics or causal mechanism produced it.

When does a "law" become vacuous?

If the variables, coding, scale, and subset are chosen only after inspecting the data, Ramsey regularity can be redescribed as a "law." Physical content requires pre-specified selection criteria, predictions beyond the fitted cases, and robustness under new observations. Ramsey's theorem guarantees existence of a pattern—not its causal significance.

DOI sources: Ramsey 1930.

Martin-Löf randomness and prefix-free incompressibility

All effective statistical tests

A Martin-Löf-random infinite bit sequence passes every algorithmically implementable test of randomness whose rejection criterion captures only an effectively vanishing fraction of fair-coin sequences. No such test singles it out as exceptional.

Algorithmic incompressibility

Descriptions are *self-delimiting*, or prefix-free, programs: no complete program begins another. For each initial segment of n bits, the *shortest* program producing it is essentially n bits long; its saving is bounded by one constant independent of n. Longer descriptions always exist.

Levin–Schnorr theorem

The two viewpoints select exactly the same infinite sequences: Martin-Löf randomness is equivalent to prefix-free incompressibility. *Algorithmic randomness* is a broader umbrella term; in this talk it means Martin-Löf randomness. No finite prefix can certify this infinite-sequence property.

DOI sources: Martin-Löf 1966; Chaitin 1975; Nies 2009.

Random sequences: almost surely, not necessarily

Single outcomes do not determine a sequence law

Value indefiniteness or irreducible chance concerns what fixes each outcome X_n. By itself, it does not specify how successive outcomes are related.

Additional postulate; exact conditional theorem

Postulate that the outcomes are independent, identically distributed fair bits. Thus, for every $s \in \{0, 1\}^n$, $\Pr\big((X_1, \ldots, X_n) = s\big) = 2^{-n}$. This is Martin-Löf's fair-coin product measure μ. For a universal test (U_m), $\mu(U_m) \leq 2^{-m}$ and the nonrandom set is $N = \bigcap_m U_m$; hence $\mu(N) = 0$ and $\mu(\mathrm{MLR}) = 1$.

Probability one $\neq$ logical necessity

The measure-one conclusion is an exact theorem *conditional on the process model*. Martin-Löf proves it for the specified computable measure; he does not prove that physical trials are independent, identically distributed fair bits. Null, nonrandom realizations still exist, and no finite prefix can certify Martin-Löf randomness.

DOI sources: Abbott et al. 2012; Martin-Löf 1966, Secs. III–IV, pp. 609–614.

Further reading

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Physical (A)Causality

Determinism, Randomness and Uncaused Events

Fundamental Theories of Physics, Vol. 192
Springer, 2018 — open access

A fuller treatment of provable unknowns, quantum and classical indeterminism, and the status of putatively uncaused events.

Thank you for your attention